

Question				Marking details	Marks available																																								
					AO1	AO2	AO3	Total	Maths	Prac																																			
6	(a)			Is a [light] {string / thread} with a {bob / mass / weight} attached to one end	1			1		1																																			
	(b)			Acceleration is proportional to its {distance / displacement} (1) and directed towards its {equilibrium / fixed} point (1)	2			2																																					
	(c)			[System A] the angle is too big [for shm to occur] (1) Any × (1) from: - the shorter length {has a larger [percentage] uncertainty / is less accurate} - the shorter period {has a larger [percentage] uncertainty / is less accurate}			2	2		2																																			
	(d)	(i)		<table><tr><th>l / m</th><th>$\sqrt{l}$ / m^½</th><th>T_{20} / s</th><th>Mean T_{20} / s</th><th>Uncertainty T_{20} / s</th></tr><tr><td>0.250</td><td>0.50</td><td>19, 21, 22, 20</td><td>21</td><td>2</td></tr><tr><td>0.500</td><td>0.71</td><td>26, 29, 26, 29</td><td>28</td><td>2</td></tr><tr><td>0.750</td><td>0.87</td><td>35, 34, 33, 36</td><td>35</td><td>2</td></tr><tr><td>1.000</td><td>1.00</td><td>42, 39, 40, 41</td><td>41</td><td>2</td></tr><tr><td>1.250</td><td>1.12</td><td>45, 42, 44, 45</td><td>44</td><td>2</td></tr><tr><td>1.500</td><td>1.22</td><td>48, 50, 49, 52</td><td>50</td><td>2</td></tr></table> [1 for each column] Deduct a maximum of 1 mark overall for inconsistent sig figs in each column	l / m	$\sqrt{l}$ / m ^½	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s	0.250	0.50	19, 21, 22, 20	21	2	0.500	0.71	26, 29, 26, 29	28	2	0.750	0.87	35, 34, 33, 36	35	2	1.000	1.00	42, 39, 40, 41	41	2	1.250	1.12	45, 42, 44, 45	44	2	1.500	1.22	48, 50, 49, 52	50	2		3		3	3	3
l / m	$\sqrt{l}$ / m ^½	T_{20} / s	Mean T_{20} / s	Uncertainty T_{20} / s																																									
0.250	0.50	19, 21, 22, 20	21	2																																									
0.500	0.71	26, 29, 26, 29	28	2																																									
0.750	0.87	35, 34, 33, 36	35	2																																									
1.000	1.00	42, 39, 40, 41	41	2																																									
1.250	1.12	45, 42, 44, 45	44	2																																									
1.500	1.22	48, 50, 49, 52	50	2																																									